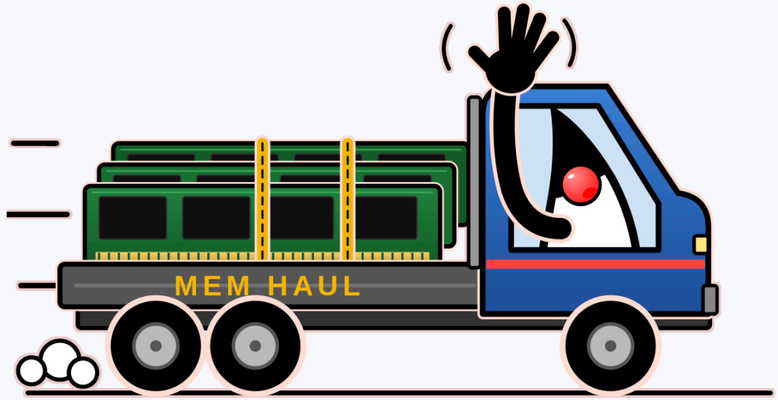
Who Moved My Heap?

Explore the Java heap for unpleasant surprises, and fix them with the right patterns, tools and runtime options.

Content and code

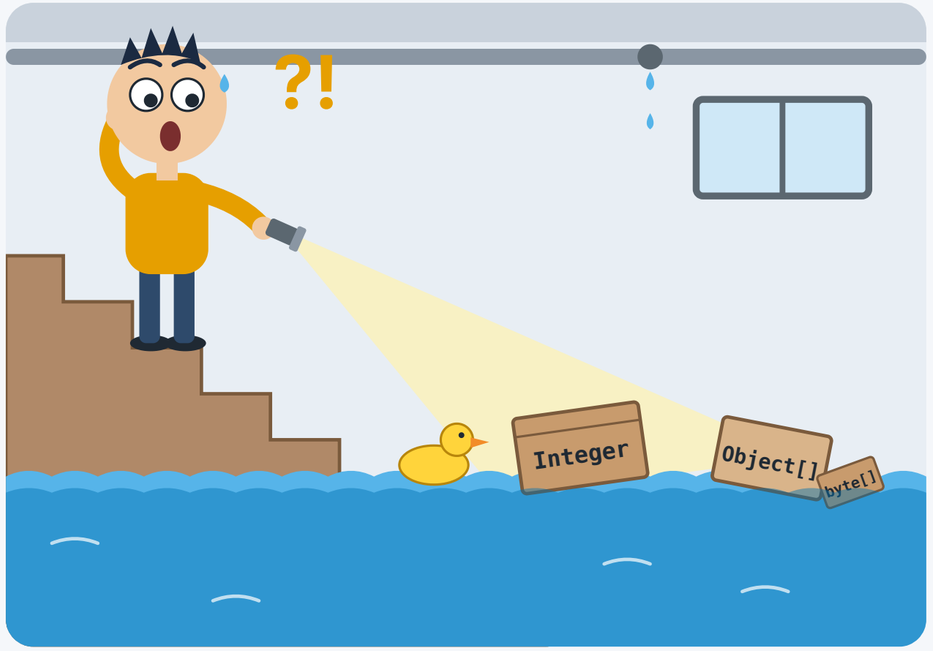
github.com/c-guntur/who-moved-my-heap



Opinions are personal and do not represent the views or opinions of the speakers' employer.

A question for you

How do you find out if your basement is flooded?



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WHO MOVED MY HEAP · 01 INTRO

Intro

A few JDK tools and a library for looking at the heap

Intro | Problem | Definitions | Solutions | AI skills | Wrap-up

01

Why look at the heap?

A few JDK tools and one library show what is on the heap of a running JVM.

- Garbage is invisible in the code, but obvious in a class histogram.
- Fewer and smaller objects mean fewer GCs and more headroom for a spike.
- `jps` finds the JVM.
- `jmap` and `jcmd` look inside it.
- JOL, a library, measures object sizes
- JMH - to measure garbage per call (covered later).

`jps`: which JVM is that?

`jps` lists the running Java processes of the current user, so you can find the pid the other tools need.

```
$ jps -l
674  Sleep.java
2310 org.apache.maven.wrapper.MavenWrapperMain
2488 jdk.jcmd/sun.tools.jps.Jps
```

- Each line is a pid and a main class or JAR.
- Adding a `-l` prints the full name.
- `-v` adds the JVM flags, which can show if the CompactObjectHeaders flag is turned on.

`jmap`: what is on the heap?

`jmap` attaches to a running JVM and prints its objects, class by class.

```
jmap -histo:live <pid>                          # reachable objects only (runs a full GC first)
jmap -histo <pid>                               # every object, garbage included
jmap -dump:live,format=b,file=heap.hprof <pid>   # full heap dump for an analyzer
```

- A histogram is cheap and small.
- A heap dump holds every object and field value.
- `:live` forces a Full GC, so insights into existing garbage are lost.

Reading a class histogram


```
 num     #instances         #bytes  class name (module)
-------------------------------------------------------
   1:         15430         698576  [B (java.base@25.0.4.1)
   2:          3302         399272  java.lang.Class (java.base@25.0.4.1)
   3:         15273         366552  java.lang.String (java.base@25.0.4.1)
```

- `[B` on the first line is a `byte[]` - a String-backed array.
- `#bytes` divided by `#instances` is the average object size.
- Comparisons between two timed histograms for a workload, can show the garbage created.

`jcmd`: one tool, many commands

`jcmd` sends a diagnostic command to a running JVM; it covers most of what `jmap` and `jinfo` did.

```
jcmd                                # list the running JVMs and their pids
jcmd <pid> help                     # list the commands that JVM understands
jcmd <pid> GC.class_histogram       # the same histogram as jmap -histo:live
jcmd <pid> GC.class_histogram -all    # includes unreachable objects, no full GC
jcmd <pid> VM.flags                 # flags in effect, e.g. +UseCompactObjectHeaders
```

- `-all` counts garbage as well, so one call before and one after shows the allocation.
- `VM.flags` proves whether compact object headers were really on.

JOL: how big is one object?

Java Object Layout (JOL) is a small OpenJDK library that shows the header, fields, padding and size of real objects.

```
List<Integer> list = new ArrayList<>(List.of(500, 501, 502));
System.out.println(GraphLayout.parseInstance(list).toFootprint());
```

```
     COUNT       AVG       SUM   DESCRIPTION
         1        32        32   [Ljava.lang.Object;
         3        16        48   java.lang.Integer
         1        24        24   java.util.ArrayList
         5                 104   (total)
```

- A library, not a JDK tool: add `org.openjdk.jol:jol-core` to the build.

Problem statement

What JDK Collections and `Stream` leave on the heap

Intro | **Problem** | Definitions | Solutions | AI skills | Wrap-up

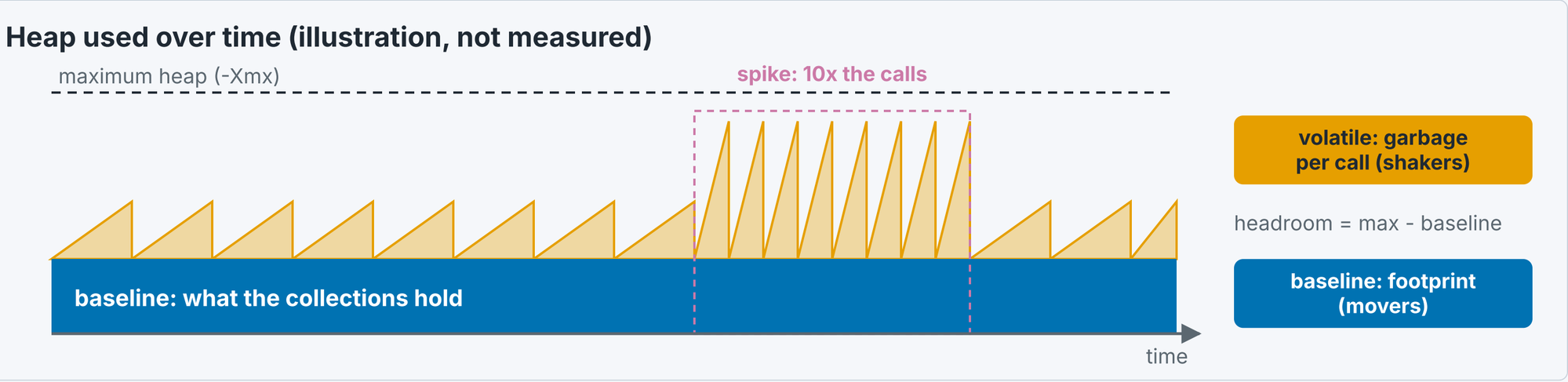
02

The problem in one slide

JDK Collections and `Stream` create objects on the heap that a developer may not know about.

- Boxed primitives and per-entry nodes inflate what a collection holds.
- Every stream call builds a pipeline of objects and drops it after.
- Amount of garbage produced depends on the effectiveness of the JVM's **Escape Analysis**.
- **Capturing lambdas** have potential to leave garbage on the heap.

Movers and Shakers



- Headroom is what is left above the baseline for garbage.
- A spike multiplies the garbage rate while the baseline stays where it was.
- **Movers** focus on baseline memory.
- **Shakers** focus on volatile memory.

Why the volatile part is the risk

Stream garbage grows with calls, not with the number of elements.

- One JDK `anyMatch` call leaves 176 B (or 160 B when using a cached `Integer`).
- At 10 M calls that is 1.76 GB of young-generation garbage.
- The same operation using Eclipse Collections allocates 0 B.

IntelliJ IDEA: what is on its heap

Top 15 of `jmap -histo` against a running IntelliJ IDEA.

```
$ jmap -histo:live 3235 | head -n 17
 num     #instances         #bytes  class name (module)
-------------------------------------------------------
   1:       3047126      214370048  [B (java.base@25.0.4)
   2:       2953580       70885920  java.lang.String (java.base@25.0.4)
   3:        239547       67348200  [I (java.base@25.0.4)
   4:        548566       44839048  [Ljava.lang.Object; (java.base@25.0.4)
   5:        176462       41384928  [J (java.base@25.0.4)
   6:        901804       28857728  java.util.HashMap$Node (java.base@25.0.4)
   7:         25466       20614184  [F (java.base@25.0.4)
   8:          6422       12487128  [C (java.base@25.0.4)
   9:         97291       12185312  java.lang.Class (java.base@25.0.4)
  10:         26398       11857104  [Ljdk.internal.vm.FillerElement; (java.base@25.0.4)
  11:         73634       11262664  [Ljava.util.HashMap$Node; (java.base@25.0.4)
  12:        457776       10986624  java.util.ArrayList (java.base@25.0.4)
  13:        328491       10511712  java.util.concurrent.ConcurrentHashMap$Node (java.base@25.0.4)
```

SonarQube for IDE: what is on its heap

Selected objects from `jmap -histo` against the SonarQube for IDE plugin process.

```
$ jmap -histo 3256 | grep "ReferencePipeline\|Lambda\|Collector\|MatchOp\|FindOp\|ReduceOp" | head -n 17
  12:        174991        9799496  java.util.stream.ReferencePipeline$Head (java.base@25.0.4)
  29:        150249        3605976  org.sonar.java.model.JSema$$Lambda/0x0000007e02e649b0
  34:         55799        3124744  java.util.stream.ReferencePipeline$3 (java.base@25.0.4)
  39:         44798        2508688  java.util.stream.ReferencePipeline$2 (java.base@25.0.4)
  41:        139439        2231024  org.sonar.java.model.JSema$$Lambda/0x0000007e02e6ca30
  44:        124810        1996960  org.sonar.python.types.v2.LazyTypeWrapper$$Lambda/0x0000007e02b08f30
  59:         55799        1339176  java.util.stream.ReferencePipeline$3$1 (java.base@25.0.4)
  72:         44798        1075152  java.util.stream.ReferencePipeline$2$1 (java.base@25.0.4)
  76:         41768        1002432  java.util.stream.MatchOps$MatchOp (java.base@25.0.4)
  77:         41711        1001064  java.util.stream.MatchOps$$Lambda/0x800000004 (java.base@25.0.4)
  78:         41711        1001064  java.util.stream.MatchOps$1MatchSink (java.base@25.0.4)
  81:         30474         975168  java.util.stream.Collectors$CollectorImpl (java.base@25.0.4)
  83:         29993         959776  java.util.stream.ReduceOps$3 (java.base@25.0.4)
  84:         29993         959776  java.util.stream.ReduceOps$3ReducingSink (java.base@25.0.4)
  99:         13687         766472  java.util.stream.ReferencePipeline$7 (java.base@25.0.4)
```

Boxing is a big cost

An `Integer` takes 16 B to hold a 4 B `int`, plus a 4 B reference to it.

1,000,000 elements, compact headers on: retained size (per element)

	JDK Boxed	EC Boxed	EC Primitive
List	20.0 MB (20 B)	20.0 MB (20 B)	4.0 MB (4 B)
Set	48.4 MB (48 B)	24.4 MB (24 B)	8.4 MB (8 B)
Map	48.4 MB (48 B)	32.8 MB (32 B)	16.8 MB (16 B)

JDK Boxed: `ArrayList<Integer>` (presized), `HashSet<Integer>`, `HashMap<Integer, Integer>`.
EC Boxed: `FastList<Integer>` (presized), `UnifiedSet<Integer>`, `UnifiedMap<Integer, Integer>`.
EC Primitive: `IntArrayList`, `IntHashSet`, `IntIntHashMap`, no `Integer` boxes.

Every `HashMap` entry is one more object

`HashMap` wraps each entry in a `HashMap$Node`.

- Each node instance is 24 B with **Compact Object Headers (COH)** on, 32 B with COH off.
- A `HashSet` contains a `HashMap`, thus also inherits the problem.
- Thus, `Node` plus `Integer` (16 B) ≈ 40 (or 48 B COH off) of heap per element to hold one `int`.

A stream is an object graph before it runs

An unevaluated pipeline already holds objects, and every operation adds a stage.

Pipeline on an empty list, compact headers on	JDK `Stream`	EC `LazyIterable`
start: `stream()` / `asLazy()`	80 B	16 B
+ `filter` / `select`	144 B	24 B
+ `map` / `collect`	144 B	24 B
+ `distinct`	128 B	16 B
3 steps: filter, map, mapToLong	272 B	96 B

Every stream call builds a pipeline and drops it

Bytes and objects per call, match at element 500, compact headers on.

Operation	JDK Object Count	JDK Stream Size	Equivalent EC Size
`anyMatch` / `anySatisfy`	7	176 B	0 B
`filter().findFirst()` / `detect`	**8**	**216 B**	**0 B**
`filter().count()` / `count`	8	224 B	0 B

- The stream's own cost is per call: the same at element 1 as at 1000.
- EC versions: capturing or fat-free, eager or lazy, all 0 B.
- **⇒ Let's dig deeper into the JDK `filter().findFirst()` to look at the 216 B allocation.**

Where the bytes come from

`list.stream().filter(p).findFirst()`, one call, match at element 500.

Origin	Objects	Bytes
Stream source: `ReferencePipeline$Head`, spliterator	2	80 B
`filter` `StatelessOp` (56 B) and its `Sink.ChainedReference` (16 B)	2	72 B
terminal op: `FindSink`, `Optional`	2	32 B
caller lambda `Integer.valueOf(test)::equals`	1	16 B
`Integer` box, outside the cache	1	16 B
Total per call	**8**	**216 B**

Problem statement

JDK Collections and Stream impact the heap.

- **Baseline**: Collections have Boxed primitives / extra classes like `HashMap$Node`.
- **Volatile**: Streams produce additional 160 B - 224 B of pipeline garbage on every stream call.
- **Volatile**: Streams do not benefit from **Escape Analysis**.

- **Statement**: Larger baselines, increased volatility risk instability if there are spikes.

Definitions

The vocabulary behind who moved my heap

03

Seven terms

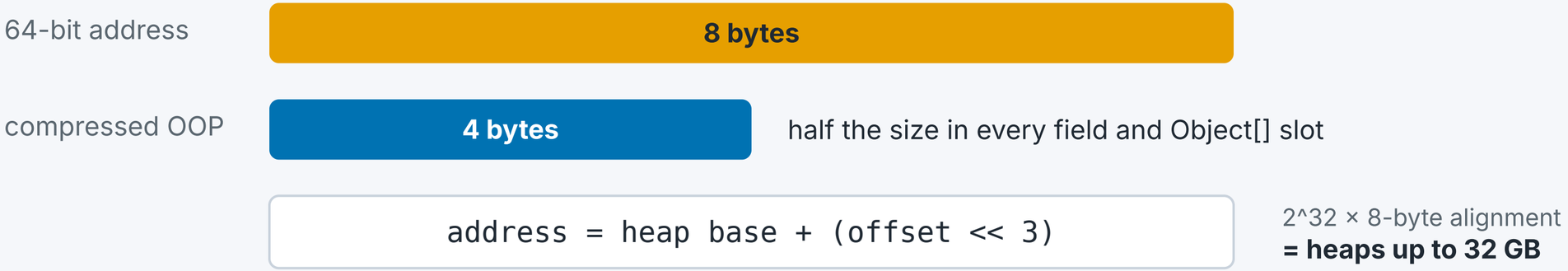
The words the solutions need, one slide each.

#	Term	In one line
01	Compressed OOPs	4-byte references on a 64-bit JVM
02	Compact Object Headers	One 8-byte header word (JDK 25+)
03	"Dollar" classes	Anonymous classes and their `Outer$1` names
04	Escape analysis	When the JIT removes *unescaped* allocations
05	Stream API	Lazy intermediate, eager terminal operations
06	Lambdas	Capturing vs fat-free
07	Immutable Data Object pooling	De-duplicating Objects

Compressed OOPs (Ordinary Object Pointer)s

For all under 32 GB heaps, Compressed OOPs produces a 4-byte offset that HotSpot stores instead of an 8-byte reference.

A reference field: uncompressed vs compressed

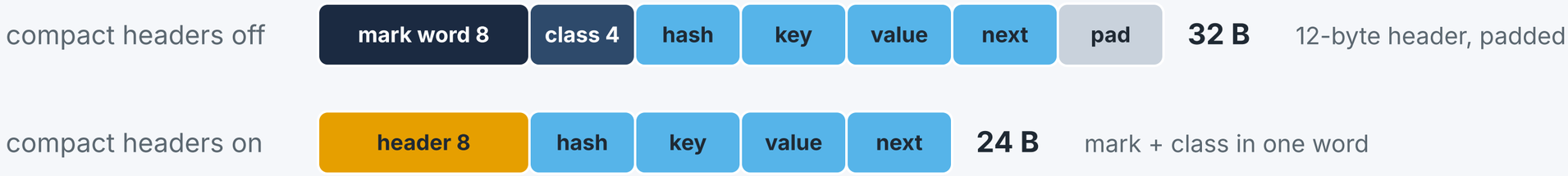


- Every reference field and `Object[]` slot shrinks from 8 to 4 bytes.
- On by default (`-XX:+UseCompressedOops`).

Compact object headers

The object header shrinks from 12 bytes to one 8-byte word (JEP 519, JDK 25).

HashMap$Node layout (box width = bytes)



1M HashMap entries: 56.4 MB → 48.4 MB. An Integer stays 16 B either way (12 B rounds up to 16).

- Off by default: `-XX:+UseCompactObjectHeaders`.
- An object only shrinks if it then fits a smaller 8-byte boundary.

"Dollar" classes

An anonymous class has no name in the code; `javac` names it `Outer$1`.

```java
public class Shaker
{
    boolean hasMatch(List<Integer> list, Integer target)
    {
        return list.stream().anyMatch(new Predicate<Integer>()  // Shaker$1
        {
            @Override
            public boolean test(Integer each)
            {
                return each.equals(target);
            }
        });
    }
}
```

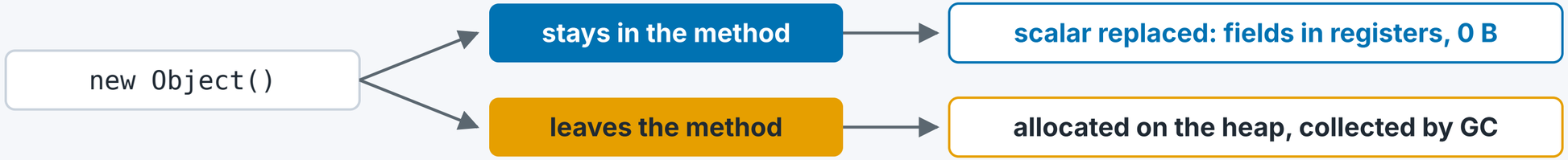
```
$ javap -p Shaker 'Shaker$1'
public class Shaker {
  public Shaker();
  boolean hasMatch(java.util.List<java.lang.Integer>,
                   java.lang.Integer);
}
class Shaker$1 implements java.util.function.Predicate<java.lang.Integer> {
  final java.lang.Integer val$target;
  Shaker$1();
  public boolean test(java.lang.Integer);
  public boolean test(java.lang.Object);
}
```

- A new instance on every evaluation, even when nothing is captured.
- Captured variables become fields: `val$target`.

Escape analysis

A C2[1] JIT[2] check: when no one outside the method can see a new object, the allocation is removed.

After inlining, C2 asks: can anyone outside this method see the new object?



Leaves the method = stored in a field, returned, or passed to code that was not inlined. Turn it off: -XX:-DoEscapeAnalysis

- Works only on compiled, inlined code, after warm-up.

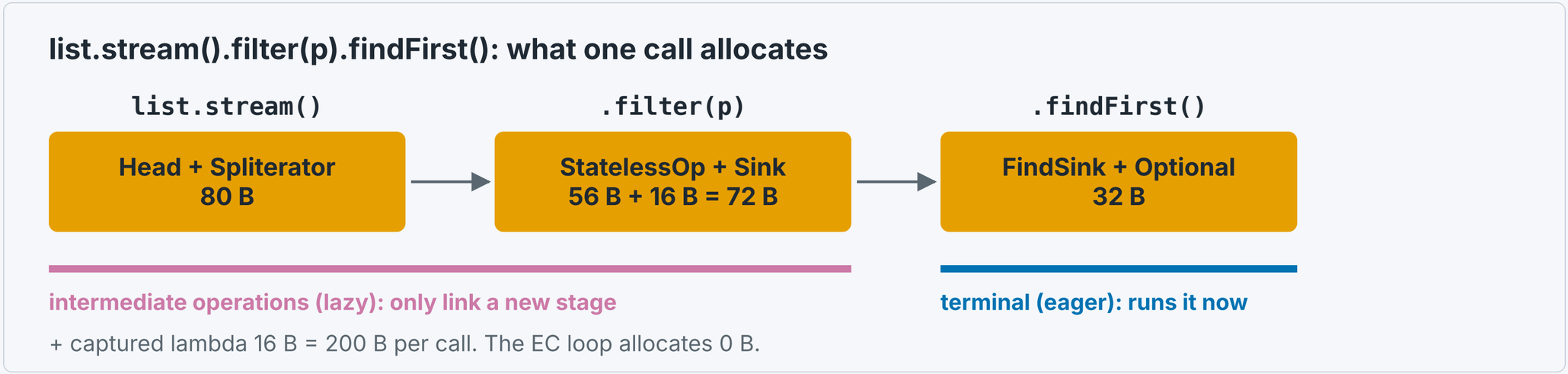
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[1] C2: the HotSpot server compiler, the optimizing compiler for hot methods (C1 is the fast client compiler).

[2] JIT: just-in-time compiler, which turns frequently run bytecode into machine code while the program runs.

Stream API: what one call allocates

Intermediate operations (`filter`, `map`) are lazy; the terminal operation (`anyMatch`, `count`) is eager and runs the pipeline.



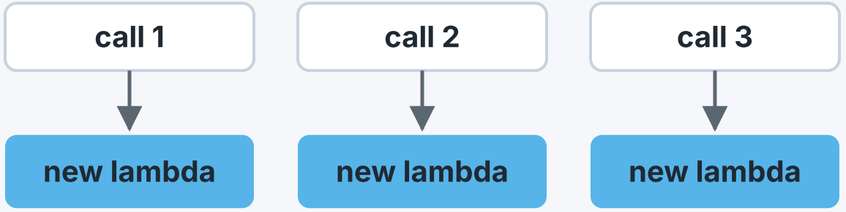
- Every call builds this graph again
- **Escape Analysis** cannot eliminate any of the produced objects.

Lambdas: capturing vs fat-free

A capturing lambda holds values from its scope. A fat-free one uses only its parameters.

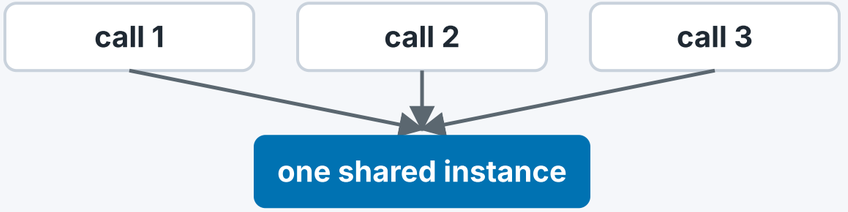
Three calls to the same method

`capturing: each -> each.equals(target)`



3 objects, each holding target

`fat-free: Object::equals, target`



target travels as an argument

- Fat-free: one instance, created once and reused on every call.
- Capturing: a new object that holds the captured values, per evaluation.

Immutable Data Object Pooling

A pool hands out one shared copy of equal immutable objects instead of many.

```java
String s1 = "Heap";                    // 1. literals are interned automatically
String s2 = "Heap";
System.out.println(s1 == s2);          // true: same object in the String pool

String s3 = new String("Heap");        // 2. explicit creation: a separate heap object
System.out.println(s1 == s3);          // false

String s4 = s3.intern();               // 3. manual interning returns the pooled copy
System.out.println(s1 == s4);          // true: same as s1
```

- JVM setting: `-XX:+UseStringDeduplication` lets the GC share the `byte[]` of equal Strings.
- A pool can be strong (entries stay until removed) or weak (the GC reclaims entries nobody uses).
- Libraries with pools: Guava `Interners`, Apache Commons, Eclipse Collections `Pool`.

Solutions

Tools, runtime options and code that move less heap

04

Solutions at a glance

Solution	Use it to
JDK CLI tooling	See what is on the heap
Java Object Layout	Size of one object Graph or Class
Java Microbenchmark Harness	Measure performance and throughput. Count bytes and objects per call
Object pooling	De-duplicate objects
Compact object headers	Shrink every header by 4 B
Non-capturing lambdas	Share one lambda instance
Eclipse Collections	Primitives, Eager & Lazy, Rich Iteration patterns

Measure first: before and after

Two class histograms around a workload show what it allocated.

```
jps -l                                                  # find <pid>
jmap -histo <pid> | grep "Lambda\|ReferencePipeline\|CollectorImpl\|HashMap\|HashSet\|ArrayList\|Integer\|Long\|Double\|Float\|Pair\|Strir
# ... run the workload ...
jmap -histo <pid> | grep "Lambda\|ReferencePipeline\|CollectorImpl\|HashMap\|HashSet\|ArrayList\|Integer\|Long\|Double\|Float\|Pair\|Strir
```

- Focus on the big ticket memory occupiers.
- Subtract per class: that is how `prof.HistoProfiler` counts objects per call.

JOL: the layout of one object

Java Object Layout (JOL) reports how the JVM lays an object out: header, fields, padding and size.

```
import org.openjdk.jol.info.ClassLayout;
import org.openjdk.jol.info.GraphLayout;

System.out.println(ClassLayout.parseClass(String.class).toPrintable());

long bytes = GraphLayout.parseInstance(array).toFootprint();
```

- `ClassLayout` shows one object while a `GraphLayout` adds up everything reachable from it.
- JOL is an OpenJDK project, added as a library. It is not part of the JDK.

JOL with compact object headers

`HashMap$Node`	Header	Fields	Size
Without CompactObjectHeaders	12 B	hash, key, value, next: 16 B	32 B (4 B padding)
With CompactObjectHeaders	8 B	the same 16 B	24 B

- Run the same JOL code with the flag off and on, and compare the sizes.
- The 4 B saved, only counts when the total drops below the next 8 B boundary.
- Note that Compact Object Headers are available from Java 25, but are not on by default.
- You can turn on Compact Object Headers with a CLI flag `-XX:+UseCompactObjectHeaders`.
- Compact Object Headers will be on by default with Java 27 and beyond.

DEMO Demo: one map entry, headers off and on


```java
import org.openjdk.jol.info.ClassLayout;

// Demo: the layout of one HashMap entry. Run it twice, compact object headers off and on.
public class NodeLayout
{
    public static void main(String[] args) throws Exception
    {
        Class<?> node = Class.forName("java.util.HashMap$Node");
        System.out.println(ClassLayout.parseClass(node).toPrintable());
    }
}
```

```
java -cp jol-core-0.17.jar -XX:-UseCompactObjectHeaders NodeLayout.java   # 32 bytes
java -cp jol-core-0.17.jar -XX:+UseCompactObjectHeaders NodeLayout.java   # 24 bytes
```

- Same fields, 4 B less header, and the 4 B alignment gap is gone.

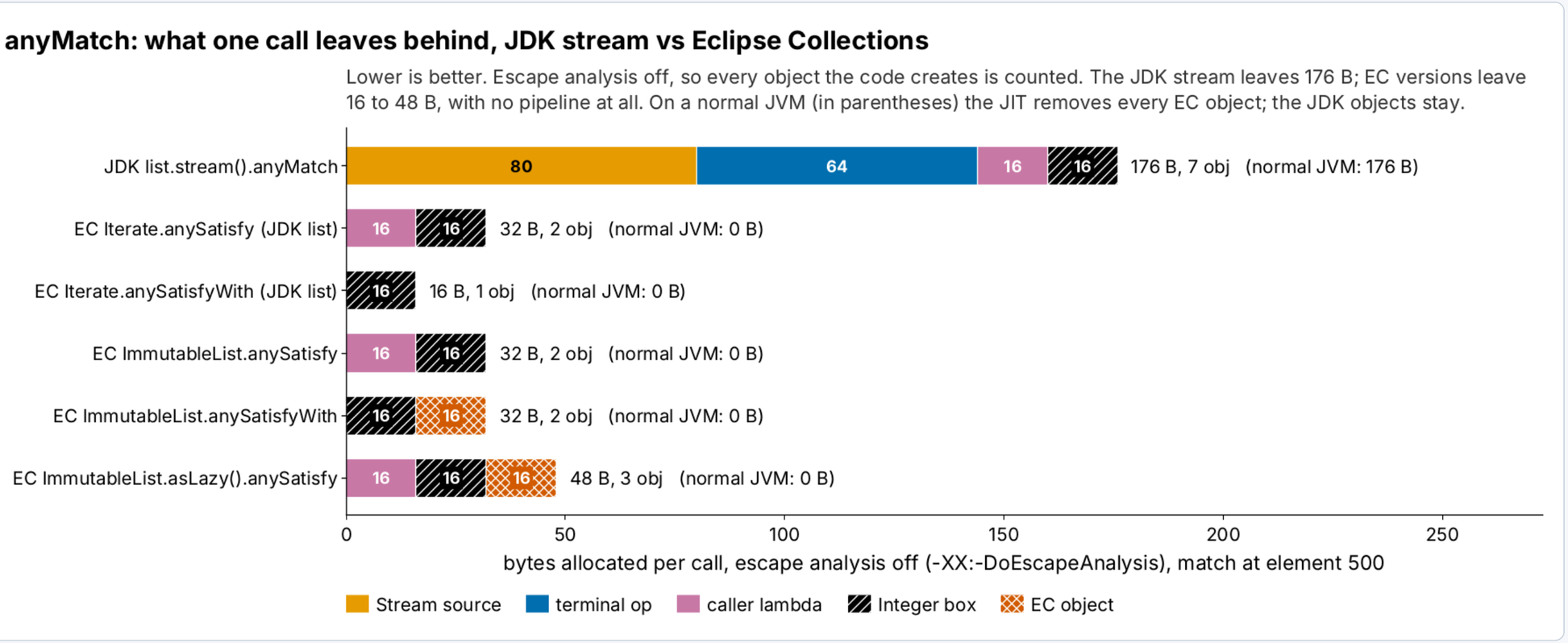
JMH: bytes per call, not only speed

anyMatch version	`return` expression
JDK list.stream().anyMatch	`jdkList.stream().anyMatch(Integer.valueOf(this.test)::equals)`
EC Iterate.anySatisfy (JDK list)	`Iterate.anySatisfy(jdkList, Integer.valueOf(this.test)::equals)`
EC Iterate.anySatisfyWith (JDK list)	`Iterate.anySatisfyWith(jdkList, Integer::equals, this.test)`
EC ImmutableList.anySatisfy	`ecList.anySatisfy(Integer.valueOf(this.test)::equals)`
EC ImmutableList.anySatisfyWith	`ecList.anySatisfyWith(Integer::equals, this.test)`
EC ImmutableList.asLazy().anySatisfy	`ecList.asLazy().anySatisfy(Integer.valueOf(this.test)::equals)`

- Source: two JMH classes in the `test` directory.
- Rows 1 to 3: `AnyMatchUtilityBenchmark`.
- Rows 1 and 4 to 6: `AnyMatchEagerLazyBenchmark`.

What one `anyMatch` call leaves behind

Escape analysis off: every object each version creates, counted.



Object pooling: de-duplicate immutable objects

Share one copy of equal immutable objects instead of allocating many.

- Integer cache: `Integer.valueOf` reuses -128 to 127; `-XX:AutoBoxCacheMax=<n>` raises the upper bound.
- String interning: `String.intern()`; `-XX:+UseStringDeduplication` shares the `byte[]` of equal Strings.
- Object interning: keep your own pool of immutable values, strong or weak.
- Libraries with pools: Guava `Interners`, Apache Commons, Eclipse Collections `Pool`.

Enabling compact object headers


```
java -XX:+UseCompactObjectHeaders -jar app.jar
jcmd <pid> VM.flags -all | grep UseCompactObjectHeaders     # check that it is on
```

Retained size	Off	On	Saved
`HashMap<Integer, Integer>`, 1 M entries	56.4 MB	48.4 MB	14%
`ArrayList<Integer>`, 1 M, presized	20.0 MB	20.0 MB	0%
`Stream.filter().map().mapToLong()`	304 B	272 B	11%

- One flag, no code change: it shrinks nodes, not boxes.

Non-capturing lambdas


```
// capturing: a new lambda on every call, holding the value
Iterate.anySatisfy(list, Integer.valueOf(test)::equals);

// fat-free: one constant lambda, the value is a parameter
Iterate.anySatisfyWith(list, Integer::equals, test);
```

- Escape analysis off: 32 B capturing, 16 B fat-free (only the `Integer` box).
- Move captured values into parameters: the `*With` methods.

Not all `anySatisfyWith()` methods in EC are non-capturing (fat-free).

Eclipse Collections: Primitives, Eager and Lazy options

	JDK Size	JDK Use	EC Size	EC Use
List of 1M `int`s	20.0 MB	`ArrayList<Integer>`	4.0 MB	`IntArrayList`
Set of 1M `int`s	48.4 MB	`HashSet<Integer>`	8.4 MB	`IntHashSet`
Match at element 500	176 B per call	`anyMatch`	0 B	`anySatisfy`
Count	224 B per call	`filter().count()`	0 B	`count`

- Primitive collections drop the need for boxed values.
- Eager methods iterate without additional garbage.
- Most of the `*With` methods keep the lambda fat-free.

AI skills

Let an assistant read the heap with you

05

AI reads a class histogram

Paste two histograms and ask what moved, not what a histogram is.

```
Here are two `jcmd GC.class_histogram -all` outputs, before and after one request.
Which classes grew most, in instances and in bytes?
Which are boxes, map nodes, arrays or lambdas, and what code likely made them?
```

- A histogram holds class names and counts; a heap dump holds your data, keep it local.
- Ask for the arithmetic, then check two rows yourself.

AI reads JOL and JMH results

Raw JMH and JOL output is long; an assistant turns it into a comparison.

- Give it the raw files (`-rf text`, JOL printouts), not a summary.
- Ask for bytes per call by variant: headers off and on, escape analysis off.
- Ask which numbers are measured and which are derived.
- Every number on these slides traces back to a raw file under `generated/runs`.

A skill to repeat the check


```
---
name: heap-check
description: Compare class histograms of a running JVM before and after a workload
---
1. Run `jps -l` and pick the pid of the app.
2. Run `jcmd <pid> GC.class_histogram -all` before and after the workload.
3. Subtract per class; list the top 10 by bytes.
4. Flag boxes, `HashMap$Node`, `Object[]` and `$$Lambda` classes.
5. Compare with the last baseline in `heap-baseline.txt`.
```

- Run it every release, not only when the basement floods.

DEMO Demo: who moved my heap?

The histogram from the first demo, pasted into an AI assistant.

```
This is `jcmd <pid> GC.class_histogram` of a running JVM.
Who moved my heap? Name the top 5 classes by bytes, say which are
boxes, nodes, arrays or lambdas, and what I could change.
```

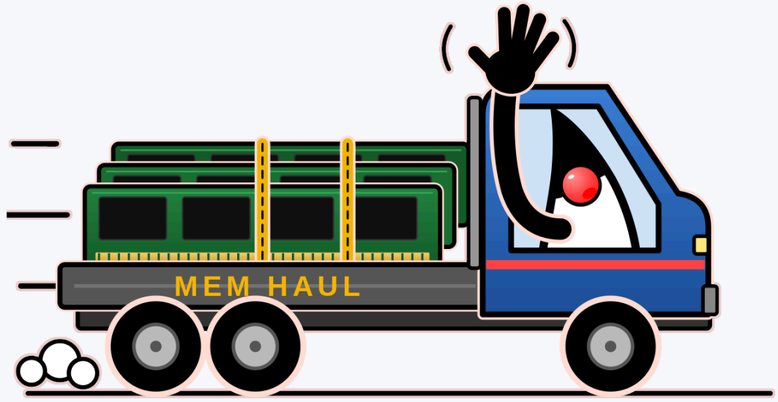
Is your basement flooded?

You only know if you go down and look.

- **Look**: `jps`, `jmap` and `jcmd`.
- **Investigate**: use JOL and JMH
- **Baseline**: primitives, pre-sized collections, compact object headers.
- **Volatile**: fat-free lambdas and loops instead of a stream per call.
- **Repeat**: a skill or a script that checks every release.

Thank you. Questions?

Content and code
github.com/c-guntur/who-moved-my-heap



Appendix

Everything that did not fit in 40 minutes

07

What is in the appendix

Appendix

#	Part	What it holds
00	Timing plan	The 40 minutes, section by section
01	Intro	`jcmd` command table, `jstat -gc` and options, more JOL output, adding JOL
02	Problem statement	Two heaps, histograms, slack, compact headers, escape analysis
03	Definitions	The full definitions: what, in code, inside the JDK
09	References	Every reference slide, by section, JOL included

Timing plan

Appendix

Start	Min	Section	Demo or audience
00:00	2	Opening	Question: is your basement flooded?
02:00	5	Intro: `jps`, `jmap`, `jcmd`, `jstat`, JOL	Demo 1: class histogram
07:00	8	Problem statement	-
15:00	9	Definitions	Show of hands: compact headers
24:00	10	Solutions	Demo 2: JOL, headers off and on
34:00	4	AI skills	Demo 3: ask who moved my heap
38:00	2	Wrap-up	Answer the opening question

`jcmd` commands for the heap

Appendix

Command	Shows
`GC.class_histogram`	Instances and bytes per class
`GC.heap_info`	Heap regions: total, committed and used
`GC.heap_dump <file>`	HPROF dump for a heap analyzer
`GC.run`	Requests a full GC
`VM.flags`	The JVM flags in effect
`VM.native_memory`	Native memory by area (needs `-XX:NativeMemoryTracking`)
`Thread.print`	Thread dump

Watching garbage with `jstat -gc`

Appendix

`-gc` shows the same counters in KB, so eden size and GC count give a rough allocation rate.

```
$ jstat -gc <pid> 1s 3                    (columns trimmed, sizes in KB)
       EC          EU          OU      YGC     YGCT
 124928.0     35840.0      2180.4      14    0.120
 125952.0    115712.0      2180.4      15    0.125
 130048.0     61440.0      7300.4      17    0.135
```

- EU climbs towards EC, a young GC runs, and eden starts again: a sawtooth.
- Eden capacity times young GCs per second is roughly the allocation rate: about 125 MB × 1.5/s ≈ 190 MB/s.
- `jstat -gccause` adds the cause of the last GC, here `G1 Evacuation Pause`.

Choosing a `jstat` option

Appendix

Option	Shows
`-gcutil`	Percent used per generation, GC counts and times
`-gc`	The same in KB: capacities and used sizes
`-gccause`	`-gcutil` plus the cause of the last and the current GC
`-gccapacity`	Minimum, maximum and current capacity per generation
`-gcnew`	Young generation: survivor spaces and tenuring
`-class`	Classes loaded and unloaded

JOL: one `Integer`, compact headers on

Appendix

With `-XX:+UseCompactObjectHeaders` the class pointer moves into the 8 B mark word.

```
java.lang.Integer object internals:
OFF  SZ   TYPE DESCRIPTION               VALUE
  0   8        (object header: mark)     0x001ac00000000001 (Lilliput)
  8   4    int Integer.value             500
 12   4        (object alignment gap)
Instance size: 16 bytes
Space losses: 0 bytes internal + 4 bytes external = 4 bytes total
```

- The header shrinks from 12 B to 8 B, but the object still rounds up to 16 B.
- The 4 B saved becomes an alignment gap: a box is a box.

JOL: the footprint of a small list

Appendix

```
List<Integer> list = new ArrayList<>(List.of(500, 501, 502));
System.out.println(GraphLayout.parseInstance(list).toFootprint());
```

Object	Count	Headers off	Headers on
`Object[]` (the backing array)	1	32 B	24 B
`Integer`	3	48 B	48 B
`ArrayList`	1	24 B	24 B
Total	5	104 B	96 B

- `GraphLayout` follows every reference, so the boxes count too: half the bytes are `Integer`.

Adding JOL to a project

Appendix

```
<dependency>
    <groupId>org.openjdk.jol</groupId>
    <artifactId>jol-core</artifactId>
    <version>0.17</version>
</dependency>
```

```
java -XX:+EnableDynamicAgentLoading -Djdk.attach.allowAttachSelf=true ...   # let JOL attach to its own JVM
java -jar jol-cli.jar internals java.util.HashMap                            # no code: the command line tool
```

- `jol-cli` is built from the openjdk/jol repository; it also has `footprint` and `heapdump-*` modes.
- jol-samples in the same repository walk through headers, alignment, arrays and more.

Two heaps in one

Appendix

A heap has a baseline that stays and a volatile part that churns.

- Baseline: the live objects your collections hold between calls.
- Volatile: short-lived garbage that each call allocates and drops.
- The baseline sets the floor; the volatile part sets how often GC runs.
- Both have to fit under `-Xmx` at the busiest moment.

Taking the histograms

Appendix

Two everyday JVMs on a developer laptop, and what they hold.

```
jps -l                          # find the pids
jmap -histo <pid> | head -15    # IntelliJ IDEA
jmap -histo <pid> | head -15    # SonarQube for IDE
```

- Without `:live` the histogram also counts garbage not yet collected.
- Look for `Integer`, `HashMap$Node`, `Object[]` and `$$Lambda` near the top.

Growing a collection leaves slack and garbage

Appendix

A list grown from empty is bigger than a presized one, and it dropped arrays on the way.

- `ArrayList` grows by 1.5x: up to 7% unused capacity in this run.
- Every array it outgrew became garbage; JOL only sees the final list.
- `HashMap` tables double: 46 to 50.5 B per entry, depending on n.
- Presize when the size is known: `new ArrayList<>(n)`, `HashMap.newHashMap(n)`.

Compact headers shrink nodes, not boxes

Appendix

`-XX:+UseCompactObjectHeaders` saves 4 B per header, but objects round up to 8 B.

Retained size	Headers off	Headers on	Saved
`HashMap<Integer, Integer>`, 1 M entries	56.4 MB	48.4 MB	14%
`HashSet<Integer>`, 1 M elements	56.4 MB	48.4 MB	14%
`ArrayList<Integer>`, 1 M, presized	20.0 MB	20.0 MB	0%
`Stream.filter().map().mapToLong()`	304 B	272 B	11%
one `Integer`	16 B	16 B	0%

Garbage you cannot read from the code

Appendix

The same line of code leaves different garbage under different conditions.

- `Integer.valueOf(1)` is cached; `Integer.valueOf(500)` costs 16 B per call.
- Escape analysis removes every EC object, and none of the stream's.
- Compact headers save 16 to 24 B per call; the objects stay.
- `anyMatch(p)` is 40 B cheaper than `filter(p).findFirst()`.

Escape analysis is not a promise

Appendix

With `-XX:-DoEscapeAnalysis` each lambda shape shows what it creates.

`anyMatch`, match at element 500	**EA on**	**EA off**
JDK `list.stream().anyMatch(…)`	176 B	176 B
EC lazy `asLazy().anySatisfy(…)`	0 B	48 B
EC capturing `Iterate.anySatisfy(…)`	0 B	32 B
EC fat-free `Iterate.anySatisfyWith(…)`	0 B	16 B

- The JIT removes allocations only in code it inlines.
- Turning it off is an upper bound, not a typical JVM.

Fat-free lambdas leave less behind

Appendix

```
// capturing: a new lambda on every call, holding the value
Iterate.anySatisfy(list, Integer.valueOf(test)::equals);

// fat-free: one constant lambda, the value is a parameter
Iterate.anySatisfyWith(list, Integer::equals, test);

// JDK stream: the pipeline is built whatever the lambda
list.stream().anyMatch(Integer.valueOf(test)::equals);
```

- Escape analysis off: 32 B capturing, 16 B fat-free, 176 B stream.
- Less garbage per call keeps the headroom free for the next spike.

What is a "dollar" class?

Appendix

An anonymous class is declared and created in one expression, without a name.

- `javac` still names it: `Outer$1`, `Outer$2`, …
- Each one gets its own `Outer$1.class` file.
- Named inner and local classes use `$` names too.

"Dollar" classes in code

Appendix

```
public class Shaker
{
    boolean hasMatch(List<Integer> list, Integer target)
    {
        return list.stream().anyMatch(new Predicate<Integer>()  // Shaker$1
        {
            @Override
            public boolean test(Integer each)
            {
                return each.equals(target);
            }
        });
    }
}
```

- Compare with lambdas (06): a non-capturing lambda is created only once.

What is object pooling?

Appendix

A pool keeps objects and hands them out again instead of allocating new ones.

- Callers return an object to the pool when they are done.
- **Worth it** for costly or limited resources: threads, connections, direct buffers.
- **Rarely worth it** for small, short-lived objects:
  - young-generation allocation is cheap;
  - pooled objects live long and need locking.

What is a compressed OOP?

Appendix

A 4-byte offset that HotSpot stores instead of an 8-byte reference.

- OOP = ordinary object pointer: a reference to a Java object.
- The offset is scaled by the 8-byte object alignment.
- On by default for heaps under 32 GB (`-XX:+UseCompressedOops`).

What is a compact object header?

Appendix

The object header shrinks from 12 bytes to one 8-byte word.

- A header holds the mark word (hash, lock, GC state) and a class pointer.
- Product feature in JDK 25 (JEP 519); experimental in JDK 24 (JEP 450).
- Off by default: `-XX:+UseCompactObjectHeaders`.

What is a stream?

Appendix

A sequence of elements processed by a pipeline: a source, intermediate operations, one terminal operation.

- Intermediate operations (`filter`, `map`) are **lazy**: they only record work.
- The terminal operation (`anyMatch`, `count`, `findFirst`) is **eager**: it runs the pipeline.
- Short-circuiting terminals stop at the first match.
- A stream can be used only once.

A pipeline in code

Appendix

```
boolean found = list.stream()                       // source
                    .filter(each -> each > 500)     // intermediate (lazy)
                    .findFirst()                    // terminal (eager)
                    .isPresent();
```

- Every call builds a new pipeline of objects.
- Nothing runs until `findFirst()`.

Inside: `ReferencePipeline`

Appendix

The internal class behind `Stream<T>`. Primitive versions: `IntPipeline`, `LongPipeline`, `DoublePipeline`.

- `stream()` creates a Head stage and a `Spliterator`.
- Each intermediate operation links one new stage (`StatelessOp` or `StatefulOp`).
- The terminal operation creates a `TerminalOp`, chains `Sink` objects and pushes elements through them.

Eager vs lazy in the JDK

Appendix

	Eager	Lazy
When work happens	At each call	At the terminal operation
In-between results	A new collection per step	None
Short-circuit	Per step	Across the whole chain
JDK examples	`anyMatch`, `count`, `findFirst`, `collect`, `removeIf`, `forEach`	`filter`, `map`, `sorted`

What is a lambda?

Appendix

An anonymous function that implements a functional interface such as `Predicate`.

- No `$` class file: `javac` emits an `invokedynamic` call.
- The runtime (`LambdaMetafactory`) builds the class on first use.
- Whether it allocates depends on what it captures.

Capturing (stateful) lambdas

Appendix

Uses a local, a parameter or `this` from the enclosing scope.

```
boolean hasMatch(List<Integer> list, Integer target)
{
    return list.stream().anyMatch(each -> each.equals(target));  // new object per call
}
```

- Each evaluation creates a new object that holds the captured values.
- Captured locals must be effectively final.

Fat-free (non-capturing) lambdas

Appendix

Uses only its own parameters, constants and static members.

```
boolean hasMatch(List<Integer> list, Integer target)
{
    return Iterate.anySatisfyWith(list, Object::equals, target);  // one shared instance
}
```

- One instance is created and then reused on every call.
- Extra values travel as arguments (the `*With` methods).

What is escape analysis?

Appendix

A C2 JIT check: after inlining, can anyone outside the method see the new object?

- **No**: the allocation is removed; its fields become locals (scalar replacement).
- **Yes**: it is allocated on the heap as usual.
- On by default; `-XX:-DoEscapeAnalysis` turns it off.

Escape analysis in code

Appendix

```
boolean hasMatch(List<Integer> list, Integer target)
{
    return Iterate.anySatisfy(list, each -> each.equals(target));
}
```

- After inlining, the lambda and the loop stay local: 0 B allocated.
- The JDK stream escapes into its own stages: 160 to 224 B per call either way.

References: Intro

Appendix

Section	Reference
jmap	The `jmap` command (Java SE 25)
jcmd	The `jcmd` command (Java SE 25)
jps	The `jps` command (Java SE 25)
jstat	The `jstat` command (Java SE 25)
jmap, jcmd, jps, jstat	Troubleshooting Guide: Diagnostic Tools (Java SE 25)
JOL	See References: JOL, the next slide

References: JOL

Appendix

Topic	Reference
JOL	OpenJDK Code Tools: Java Object Layout (JOL)
Source, CLI	openjdk/jol: jol-core, jol-cli and jol-samples
Samples	jol-samples: headers, alignment, arrays and more
Library	Maven Central: `org.openjdk.jol:jol-core`
Headers	JEP 450: Compact Object Headers (Experimental), JDK 24
Headers	JEP 519: Compact Object Headers, JDK 25
Headers	OpenJDK Project Lilliput
Flag	The `java` command (Java SE 25): `-XX:+UseCompactObjectHeaders`

References: Baseline and volatile heap

Appendix

- HotSpot Virtual Machine Garbage Collection Tuning Guide (Java SE 25)
- Factors Affecting Garbage Collection Performance (Java SE 25)
- The `java` command: `-Xmx` and GC options (Java SE 25)

References: jmap in the wild

Appendix

- The `jmap` command (Java SE 25)
- The `jps` command (Java SE 25)
- Troubleshooting Guide: Diagnostic Tools (Java SE 25)

References: Movers

Appendix

- Java Object Layout (JOL), OpenJDK Code Tools
- `Integer.valueOf(int)` (Java SE 25 API)
- `HashMap` (Java SE 25 API)
- `ArrayList` (Java SE 25 API)
- The `java` command: `-XX:+UseCompactObjectHeaders` (Java SE 25)

References: Shakers

Appendix

- Package `java.util.stream` (Java SE 25 API)
- The Java Tutorials: Lambda Expressions
- Java HotSpot VM Performance Enhancements: Escape Analysis (Java SE 25)
- Java Magazine: Escape analysis in the HotSpot JIT compiler
- HotSpot Virtual Machine Garbage Collection Tuning Guide (Java SE 25)

References: The problem

Appendix

- HotSpot Virtual Machine Garbage Collection Tuning Guide (Java SE 25)
- Package `java.util.stream` (Java SE 25 API)
- The `jmap` command (Java SE 25)
- Java Object Layout (JOL), OpenJDK Code Tools

References: "Dollar" classes

Appendix

- The Java Tutorials: Anonymous Classes
- JLS SE 25, 15.9.5: Anonymous Class Declarations
- JLS SE 25, 13.1: The Form of a Binary ($ names)
- JLS SE 25, 8.1.3: Inner Classes and Enclosing Instances

References: Object pooling

Appendix

- `Integer.valueOf(int)` (Java SE 25 API): the -128 to 127 cache
- JLS SE 25, 5.1.7: Boxing Conversion
- The Java Tutorials: Thread Pools
- Java Garbage Collection Basics: why short-lived objects are cheap

References: Compressed OOPs

Appendix

- Java HotSpot VM Performance Enhancements (Java SE 25): Compressed Ordinary Object Pointer
- The `java` command (Java SE 25): `-XX:+UseCompressedOops`

References: Compact object headers

Appendix

- JDK 25 Release Notes: JEP 519 Compact Object Headers
- The Arrival of Java 25 (Oracle Java blog)
- The `java` command (Java SE 25)

References: Stream API

Appendix

- `java.util.stream` package summary (Java SE 25): pipelines, laziness, low-level construction
- `Stream` (Java SE 25 API)
- The Java Tutorials: Aggregate Operations
- The Java Tutorials: Reduction

References: Lambdas

Appendix

- The Java Tutorials: Lambda Expressions
- JLS SE 25, 15.27: Lambda Expressions (15.27.2 effectively final, 15.27.4 evaluation)
- `LambdaMetafactory` (Java SE 25 API)
- Java 8: Lambdas, Part 1 (Oracle technical article)

References: Escape analysis

Appendix

- Java HotSpot VM Performance Enhancements (Java SE 25): Escape Analysis
- Escape Analysis in the HotSpot JIT Compiler (Oracle Java Magazine)
- The `java` command (Java SE 25): `-XX:+DoEscapeAnalysis`

References: Solutions

Appendix

- OpenJDK Code Tools: Java Microbenchmark Harness (JMH)
- OpenJDK Code Tools: Java Object Layout (JOL)
- JEP 519: Compact Object Headers
- The `jcmd` command (Java SE 25)
- The Java Tutorials: Thread Pools
- Eclipse Collections